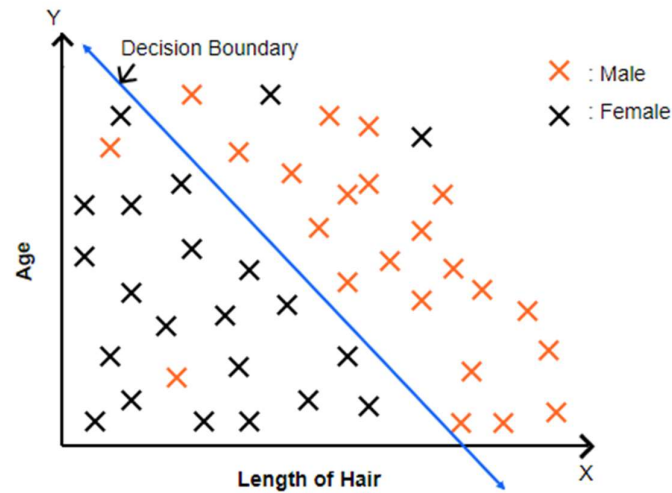


Logistic Regression



Logistic Regression is a supervised machine learning algorithm for classification, usually binary (0 or 1). Instead of predicting a continuous value, it predicts the probability of a class:

$$0 \leq \hat{y} \leq 1$$

We have features $f_1, f_2, \dots, f_d$ (let's denote as vector f) and parameters w (weight vector).

The linear combination of features is:

$$z = w_0 + w_1 f_1 + w_2 f_2 + \dots + w_d f_d = w^T f$$

We map z to a probability using the sigmoid function:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

So, the predicted probability that $y = 1$ is:

$$\hat{y} = P(y = 1 | f; w) = \sigma(w^T f)$$

Cost Function (Binary Cross-Entropy)

The loss for one example is:

$$L(w) = -[y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})]$$

For n examples, the cost function $J(w)$ is:

$$J(w) = -\frac{1}{n} \sum_{i=1}^n \left[y^{(i)} \log(\hat{y}^{(i)}) + (1 - y^{(i)}) \log(1 - \hat{y}^{(i)}) \right]$$

Gradient of the Cost Function

The gradient w.r.t weight w is:

$$\frac{\partial J(w)}{\partial w} = \frac{1}{n} \sum_{i=1}^n (\hat{y}^{(i)} - y^{(i)}) f^{(i)}$$

- In vectorized form:

$$\hat{y} = \sigma(FW) \quad (F: n \times d \text{ matrix}, W: d \times 1 \text{ vector})$$

$$\nabla J(W) = \frac{1}{n} F^T (\hat{y} - y)$$

Gradient Descent Update

$$w := w - \alpha \nabla J(w)$$

Decision Boundary

After training, to classify:

$$\text{if } \hat{y} \geq 0.5 \text{ then } y = 1, \text{ else } y = 0$$

- The boundary is linear in f : $w^T f = 0$.